	space.	
3a	the *increase* in internal energy of a system is equal to the sum of the *heat supplied to* the system and the *work done on* the system. Only get 1 mark if candidate did not use the word "system".	A1
3b	At constant volume, no work is done by the gas. Heat required is solely used to increase its internal energy. At constant pressure, work is done by the gas as it expands. Heat is required for the increase of its internal energy and the work done by the gas. More heat is required to increase the temperature of unit mass of gas by one unit for constant pressure. From $Q = mc\Delta\theta$, more heat implies larger specific heat capacity.	B1 B1 A0
3c	Internal energy consists of potential energy due to intermolecular forces and kinetic energy due to random motion. Temperature is only making reference to the kinetic energy due to random motion. Temperature is proportional to the average kinetic energy due to random motion but internal energy also depends of the total number of particles in the system.	B1 B1

3di1	Equation of state: $pV = nRT$ $2.62\times10^{5}(0.0360) = n(8.31)(273.15+25)$ $n = 3.8069 \text{ mol} = 3.8 \text{ mol}$	M1
3di2	$pV = nRT$ $p(0.0360) = (0.30+3.8069)(8.31)(273.15+25)$ $p = 2.8217\times10^{5} \text{ Pa}$	C1 A1

4a	It is a wave in which the oscillations of the particles in the wave are along the direction	B1